

This report presents a statistical analysis of the 2025 Stack Overflow Developer Survey (N=20,407) to investigate two primary objectives: 1) distinguishing the economic implications of remote versus hybrid job modes in the technology sector, and 2) evaluating the effect of education levels on income on North America (USA and Canada).

Statistical summaries of the raw data revealed that the target variable, *ConvertedCompYearly* (salary), is heavily right-skewed with a mean of \$102,117 and a standard deviation of approximately \$340,000, driven by extreme outliers (max > \$33.5M). Similarly, the Age variable displayed data entry errors, with values ranging from 1 to 151 years, necessitating filtering for realistic demographic analysis.

To further understand the underlying structure of the data, three specific relationships were visualized and diagnosed for anomalies. Figure 1 reveals an interesting anomaly that "Student" ranks third. Their satisfaction score outranks many professional engineering roles, which may indicate that the learning phase is often more satisfying than the pressures of professional employment. Figure 2 indicates that organizational responsibility is a stronger motive for compensation than technical expertise alone, since Senior Executives and Engineering Managers distinctly outrank specialized technical roles, like AI or Cloud engineers. Figure 3 shows a distinct heteroscedastic relationship, where salary variance expands significantly with age, indicating that while experience unlocks higher earning ceilings, it does not guarantee a higher floor, particularly as the trend plateaus after age 50.

To address the first objective, the analysis narrows its scope to specifically examine the salary distinctions between Hybrid and Remote job modes. The maximum salaries are extremely large (\$18.4M for Hybrid and \$33.6M for Remote) compared with the medians (\$77,730 and \$90,527). The minimum values (around \$100) also look unrealistic and may reflect data entry issues or non-comparable work situations. Because the later two-sample t-test is based on the mean and variance, these extreme values can strongly affect the results, making it harder to detect a real difference between groups. Therefore, the data was trimmed by removing values outside the 1st (\$848) and 99th (\$450,000) percentiles of the combined dataset. This refinement restricted the analysis to a representative salary range of approximately \$850 to \$450,000, reducing the sample size slightly (Hybrid: 7,681; Remote: 6,764) but ensuring robust statistical validity.

To compare the mean salaries, a manual Student's t-test was initially calculated assuming homogeneity of variances (Pooled SD = 69,594), $t(14443) = -13.13$. However, consistent with the heteroscedasticity observed in the EDA, Levene's test confirmed a statistically significant difference in variances between the Remote and Hybrid groups ($p < .001$). Consequently, the assumption of equal variance was rejected in favor of Welch's t-test. This method corrected unequal variability and produced a slightly adjusted statistic of $t(14443) = -12.98$, $p < .001$. These parametric results suggest that Remote employees (M=104,784, SD=75,873) earn significantly more than their Hybrid employees (M=89,552, SD=63,554), with a mean difference of approximately \$15,231.

To validate these findings against potential non-normality, a non-parametric bootstrap analysis (B=10,000) was performed. Figure 4 reveals that the bootstrapped mean distributions for Hybrid and Remote are distinct and non-overlapping, signifying a clear salary hierarchy that continues across repeated resampling. The distribution of their difference (Fig 5) is centered at -\$15,218 (SE=1,174) and entirely excludes zero, while the symmetric, near-normal standardized distribution in Figure 6 justifies the parametric approach used earlier. Based on the analysis, the observed income disparity between work modes is statistically robust and independent of specific distributional assumptions.

Recalculating the t-statistic using the bootstrapped standard error suggests $t = -12.97$, almost identical to the Welch's $t (-12.98)$ reported earlier. This tiny discrepancy suggests the robustness of the

Welch's test against non-normality. In contrast, the larger difference of the pooled $t(-13.13)$ aligns with the Levene's test violation ($p < .001$). The comparison of different t tests illustrate that the violation of the equal-variance assumption caused the pooled method to underestimate the standard error, thereby artificially inflating the absolute t -value.

While the bootstrap-based t -test indicates statistical significance, confidence intervals provide an estimate of the magnitude of the mean difference and the associated uncertainty. All three calculated 95% CIs for the mean salary difference (Hybrid – Remote) entirely exclude zero. This result confirms a statistically significant disparity in the average salary between Hybrid and Remote groups. Using the original data, the Welch-based CI was $[-\$17,532, -\$12,931]$ ($df \approx 13,248$), while the pooled t -based CI was $[-\$17,506, -\$12,957]$ ($df = 14,443$). The bootstrap percentile CI ($B=10,000$) shows a highly consistent range of $[-\$17,483, -\$12,923]$. These intervals are strictly negative, which suggests that Hybrid workers earn significantly less on average than Remote workers. The strong alignment between the Welch and bootstrap intervals suggests the inference is robust to distributional concerns. Although the pooled CI is still similar, its vulnerability to the questionable assumption of equal variance, which was already disproven by the Levene test, makes the Welch and bootstrap methods more preferable from a theoretical standpoint.

Beyond comparing means, a Mood's median test was conducted to evaluate the central tendency across the entire distribution. With a grand median of \$81,920, the test revealed a strong association between job mode and salary level relative to the median, $\chi^2(1) = 112.98, p < .001$, effectively rejecting the null hypothesis of equal medians. The observed median difference (Hybrid minus Remote) was $-\$13,064$, which was closely mirrored by a bootstrap analysis ($B=10,000$), with a mean median difference of $-\$13,249$ ($SD=1,582.1$). The bootstrap-based p -value ($p < .0001$) further confirms the robustness of this disparity. Collectively, these findings reinforce the conclusion that Hybrid workers consistently earn less than Remote workers at the median level in this dataset.

To address the second objective, the analysis below identified that the outcome variable was salary, and the grouping variable was education level with three categories, Bachelor's, Master's, and Professional degree. Initial data exploration showed the presence of extreme salary values; therefore, salaries were winsorized at the 1st and 99th percentiles rather than trimmed since one-way ANOVA can be sensitive to unequal variances. This approach retained the full sample size ($n_{\text{Bach}}=2,978$, $n_{\text{master}}=1,084$, $n_{\text{Prof}}=301$) while stabilizing within-group variance, making ANOVA assumptions more robust. The winsorized mean salaries were \$158,256.95 for the Bachelor's group, \$176,813.24 for the Master's group, and \$185,269.16 for the Professional group.

A one-way ANOVA indicated a significant effect of education level on mean salary, $F(2, 4360) = 21.31, p < .001$. This result rejects the null hypothesis that all three education groups have the same mean salary in the North American subsample. To assess robustness without relying on normality, a nonparametric bootstrap procedure with 10,000 replications was conducted using sample sizes matched to each group. The bootstrap results were consistent with the ANOVA pattern. The average bootstrap mean differences were $-\$18,626.64$ (Bachelor's minus Master's), $-\$26,911.63$ (Bachelor's minus Professional), and $-\$8,284.99$ (Master's minus Professional). These estimates indicate that the mean salary increases through advanced degrees in this sample. Finally, a bootstrap-based ANOVA test was also performed by generating a null distribution of the F statistic. The observed F statistic ($F=21.31, p < .0001$) matched the classical ANOVA. Both classical and bootstrap ANOVA suggest that mean salaries differ across the three education groups for North American respondents, and advanced degrees provide a statistically robust income premium.

In conclusion, this current multi-methodological analysis demonstrates that both remote work and higher educational attainment are positively related to significant salary gains in the technology industry.

These results have been shown to be statistically significant using both parametric tests and non-parametric bootstrap simulations.

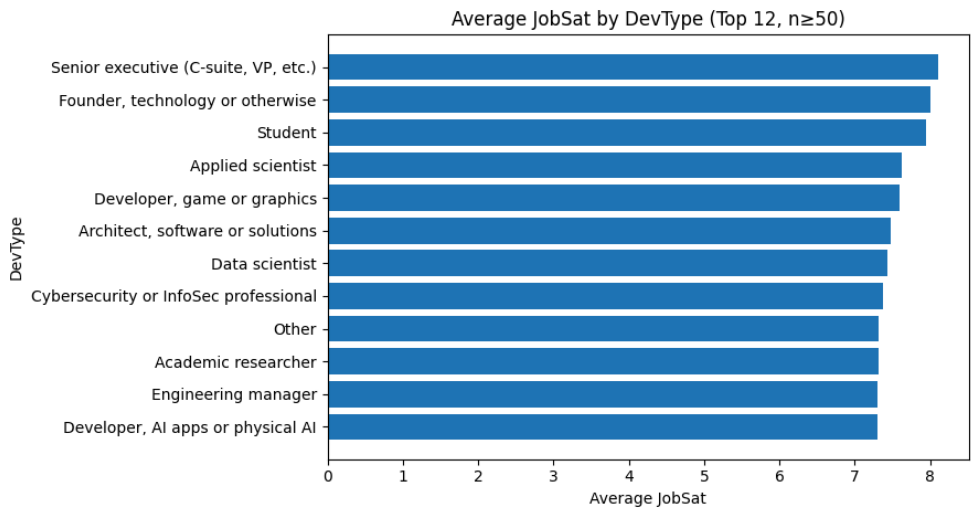


Fig 1. Average Job Satisfaction by Job Title (DevType)

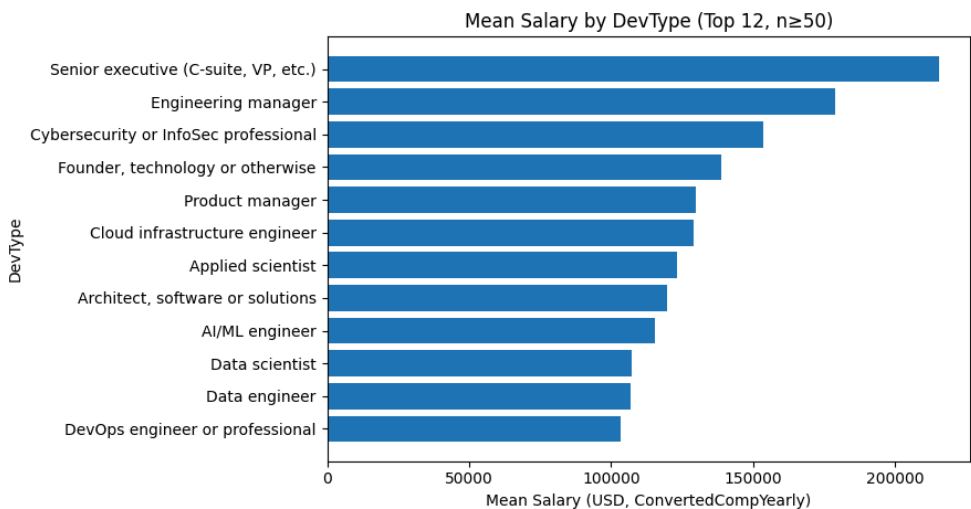


Fig 2. Mean Salary by Job Title



Fig 3. Annual Salary vs. Age

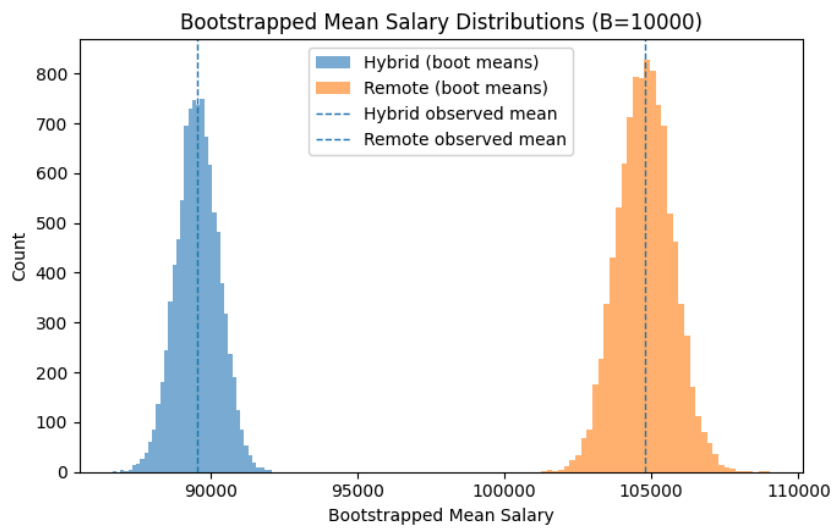


Fig 4. Bootstrapped Mean Salary Distributions

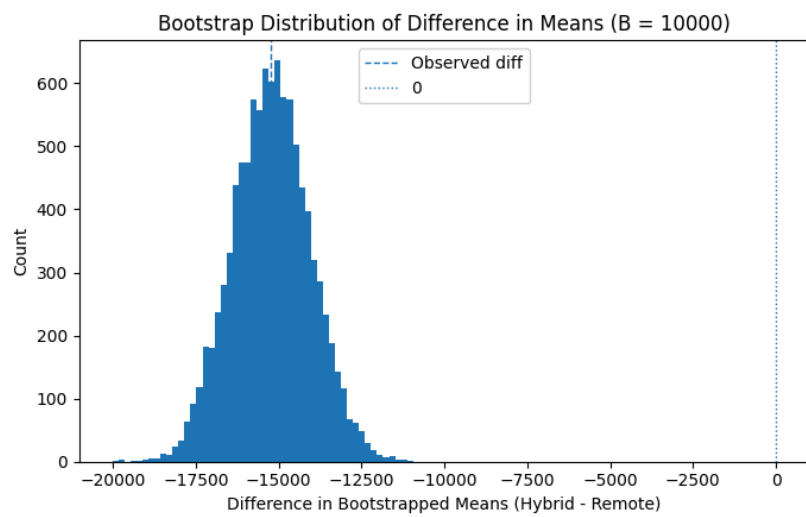


Fig 5. Bootstrap Distribution of Difference in Means

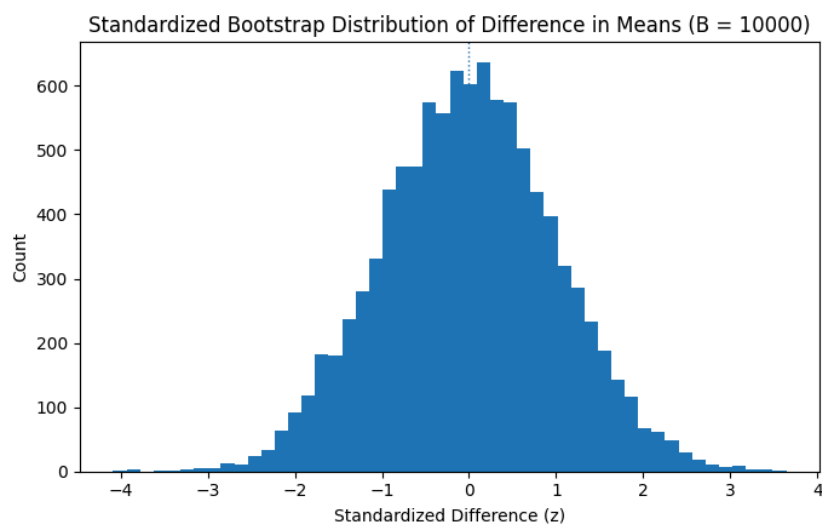


Fig 6. Standardized Bootstrap Distribution of Difference in Means

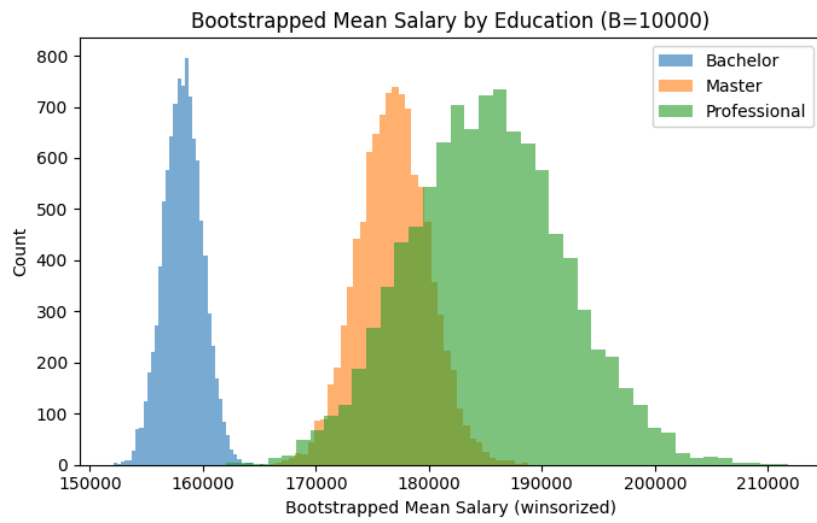


Fig 7. Bootstrapped Mean Salary by Education

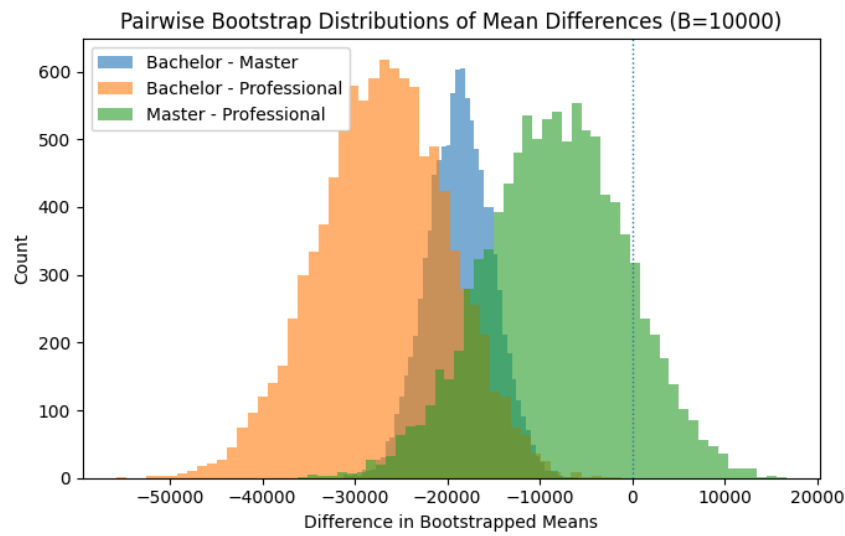


Fig 8. Pairwise Bootstrap Distributions of Mean Differences

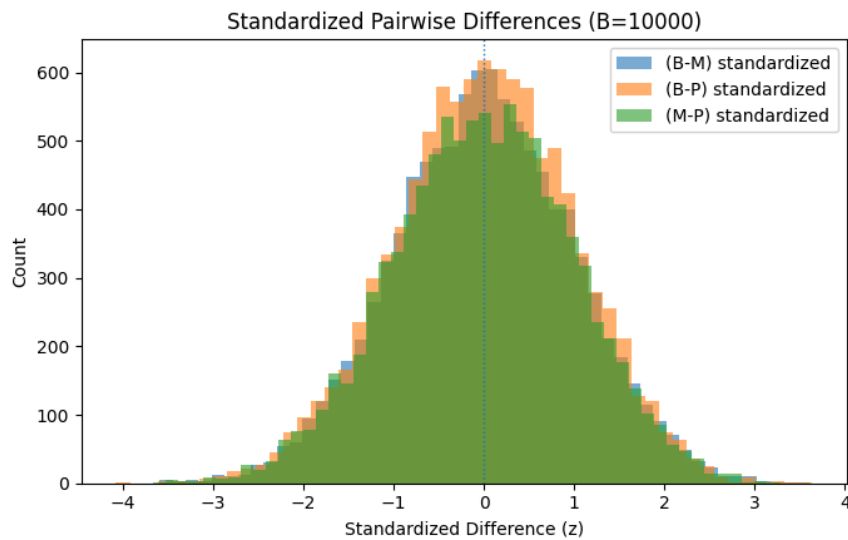


Fig 9. Standardized Pairwise Differences

Appendix

Statement of LLM Usage

LLM was used to support the interpretation and formatting (general questions) in this report. Specifically, the LLM was used to help me with learning the statistical concepts including bootstrap inference, Welch's correction under unequal variances, and interpretation of small differences between a bootstrap-based t statistic and the Welch t statistic and to improve wording for reporting results.

Example prompts included: "could you explain me a bit more about why the t-statistic recalculated from the bootstrapped standard error might slightly differ from the Welch's t-statistic. Does this discrepancy indicate a failure of parametric assumptions, or is it a natural variation of resampling?", "how to write a square symbol in google doc?", and "I think we should use Welch's correction for violating the equal-variance assumption."

A Specific Example of encountering error with LLM

During Question 3, where I had to handle the outliers, I think using trimming techniques like in Question 2 would be unreasonable for Question 3 since I realized this subsample from North America is smaller than the global sample used in Question 2. So I asked my LLM this question, like 'What are some suggestions you would provide for the outlier handling?' LLM's answer was trimming, and I sent out my argument (Anova is very sensitive to unequal variances and influential outliers. Also, if we use trimming there will be about 200 data deleted from the subsample). Then, my LLM considered my argument valid, so I ended up using winsorization. If I did not have this presumption and asked directly for an answer without any critical thinking, I might be losing 200 data and the analysis to Question 3 could be less convincing.